

Digital Payment Ecosystems

Claude Shannon’s Fraud Catchers | September 2026

Every figure regenerates from the public repository (<https://github.com/kimmeyh/hsbc-quantum-fraud-2026>, freeze commit 95751b9, tag `prereg-freeze`) and carries a configuration hash and an evidence tag: **[HW]** measured on Dirac-3, **[SIM]** simulated or classical, **[PROJ]** projected. The repository holds the resolved environment (experiments/requirements-lock.txt: eqc-models 0.21.0, qci-client 5.0.2, Python 3.12, Linux), both dataset checksums, the preregistration, all amendments, the results store, sixty QCi job identifiers, 48 raw device responses and the Dirac-3 parameters; the A21 probe has no identifier and no reported figure depends on it.

A. Results

A.1 ULB detection quality. 284,807 transactions, 1,081 exact duplicates removed before splitting, 60/20/20 stratified, seeds 42 to 51, test-fold prevalence 0.00167. AUPRC is step-wise average precision throughout; intervals are t-intervals across ten overlapping resplits and describe split sensitivity, not sampling error.

Arm	Features	AUPRC	Seed SD	95% interval	Tag
CatBoost	30	0.8368	0.0304	0.8150 to 0.8585	SIM
XGBoost	30	0.8296	0.0286	0.8092 to 0.8501	SIM
CatBoost (H1b comparator)	13	0.8070	0.0321	0.7841 to 0.8300	SIM
CVQBoost on Dirac-3 (frozen, 91 variables)	13	0.7671	0.0302	0.7455 to 0.7887	HW
Logistic regression	30	0.7218	0.0268	0.7026 to 0.7410	SIM

The exact classical solve of the hardware row scores 0.7681; a tuned 9-feature pool scores 0.7014 [HW]. Every ULB configuration except B2 uses singles plus pairs ($13 + 78 = 91$ variables).

A.2 Operational view. About 95 frauds in 56,746 test transactions: the 0.05% and 0.1% budgets fund 28 and 57 alerts, capping recall at 0.295 and 0.600; only 0.5% is uncapped. Proxy figures are [SIM] because the first campaign did not persist Dirac-3 weights.

Arm	R@0.05% (cap 0.295)	R@0.1% (cap 0.600)	R@0.5%
CatBoost, 30 features	0.294 (99.7% of cap)	0.593 (98.8%)	0.855
CatBoost, 13 features	0.286 (96.9%)	0.580 (96.7%)	0.849
CVQBoost proxy, 13 features [SIM]	0.283 (95.9%)	0.565 (94.2%)	0.819
CVQBoost mixed pool, 6 features [HW]	0.271	0.509	0.839
Logistic regression	0.241 (81.7%)	0.513 (85.5%)	0.839

A.3 Hardware campaign. 61 metered fits, 1,141 device seconds, zero failures, zero retries, 4 to 92 s per fit; every fit ran `relaxation_schedule 2`, `sum_constraint 1`, `num_samples 8`. G0b: Spearman 0.900 over five paired configurations (exact permutation p 0.042 one-sided), a smoke test only. H1b: -0.0399 [-0.0571, -0.0227], nine of ten seeds negative [HW]; six of ten per-seed paired BCa intervals exclude zero [SIM]. Within-fit energy spread across eight draws: median 0.020%, maximum 0.343% over 48 fits [HW].

A.4 Solver fidelity and mechanism controls. Hardware minus exact proxy on identical Hamiltonians: -0.0010 [-0.0032, +0.0012] AUPRC; weight cosine 0.975 to 0.999 on order-2 pools (the order-3 pool reaches 0.83, B.3); hardware objective 0.013% to 0.413% above the exact minimum, never below [HW], which holds by construction against a certified convex minimum. On the frozen pool the agreement is forced rather than earned. QCi documents an effective analog resolution of about 200:1 (Emami et al. cite a 23 dB dynamic-range limit). The frozen Hamiltonian’s diagonal is 510,705 while its off-diagonal entries differ by at most 20.0 and its linear terms by at most 28.0, against a resolvable difference of 2,554: no coefficient difference is visible to the device, the quantized problem’s minimizer is uniform to $2e-15$, and hardware returning uniform is what resolution predicts whatever the true optimum is (A31).

The frozen optimum is uniform to within $3.3e-06$ to $7.4e-06$ in L1 of $1/91$ on all ten seeds, under a solve certified by KKT residual below $1e-9$ (A26). Classical controls at zero metered cost: uniform weights 0.7659, solved 0.7687, class-weighted objective 0.7690, free-sign logistic stack 0.7681, so the simplex form costs nothing detectable and objective reweighting is ruled out. The +0.0028 is tie-breaking: 95.2% of test rows share one score and reordering within that block spans 0.0120 of AP. Cause: the unbounded trees memorise the 170,235-row fold; 80 to 84 of 91 learners reproduce the labels exactly, and zero predict the negative class everywhere

(A20). At $\lambda = 0$ the minimiser set is a face of the simplex, so the penalty selects a diffuse solution near uniform rather than being irrelevant to it (A27).

Mixed pool (exploratory, A11). Four families at $k=13$, 312 variables, same splits: L1 from uniform 0.126, solved minus uniform +0.0076 on 10 of 10 seeds, absolute AUPRC 0.7466, below the frozen pool’s 0.7681 [SIM]. *Tuned pool (A13, A14, A16)*. Fit-time class weighting in the tree and logistic learners (LDA and KNN accept none) plus closer distance-weighted neighbours: Gram ratio falls from 0.9988 to 0.92; selected on validation AP at seed 42, then ten seeds: AUPRC 0.7700 (SD 0.0270); against the single-family pool rebuilt at the same $k=6$ on the same splits, +0.0319 (SD 0.0189), 10 of 10 seeds; solved minus uniform +0.0047 (SD 0.0028). *Hardware block*. The ceiling foreclosed $k=13$ at the time, so the metered mixed-pool block used $k=6$, 60 variables: ten fits, 43.0 device seconds, AUPRC 0.7630, hardware minus proxy -0.0007, cosine 0.977 to 0.983 [HW]; it measures solver fidelity, not the $k=13$ mechanism.

A.5 IEEE-CIS. 590,540 transactions, 3 duplicates removed, 3.5% prevalence, rolling-origin evaluation over 30-day TransactionDT buckets, reduced feature recipe after Deotte with UID excluded. Evaluation periods hold 85,302 / 86,524 / 8,111 rows; headline figures are unweighted means and the brackets below are fold min to max across the three folds, not confidence intervals. The shuffled-label control (permuted training labels, true evaluation labels) lands at or below the base rate on every fold; a leaking pipeline would score above it. Hyperparameters are carried from ULB as a hypothesis, so classical figures are floors.

Arm	Mean AUPRC	Features
LightGBM	0.5739 [0.5424, 0.6293]	~182
XGBoost	0.5028 [0.4689, 0.5260]	~182
CatBoost	0.4795 [0.4696, 0.4927]	~182
CVQBoost tuned / frozen [SIM]	0.0571 / 0.0523	6

Matched-feature control. LightGBM restricted to the same six features falls from 0.5424 to 0.0734; CVQBoost attains 85% of that ceiling (0.0621 of 0.0734, fold 0). Gram ratios 0.950 to 0.973 rule out the A.4 degeneracy mode. *H3 ladder (12 cells, proxy)*. CVQBoost minus matched GBDT: -0.0169 ($k=5$), -0.0764 ($k=9$), -0.0564 ($k=13$), -0.1031 ($k=17$); slope -0.006 per feature. *The same ladder on hardware (A22)*. Same pipeline, folds and 100,000-row pool subsample; only the solver differs. Twelve fits, 62 metered seconds, pairs only:

k	Variables	AUPRC [HW]	AUPRC [SIM]	Difference
5	10	0.0510	0.0508	+0.0002
9	36	0.0804	0.0806	-0.0002
13	78	0.1172	0.1174	-0.0002
17	136	0.1430	0.1427	+0.0003

Means over three folds at prevalence 0.0368; the largest cell discrepancy is 0.0013 and errors scatter both ways (five above, five below, two exact). CVQBoost trails the matched GBDT at every rung (-0.1028 [HW] at $k=17$), so the gap belongs to the formulation. An earlier hardware ladder that appeared to triple CVQBoost’s AUPRC had built pools on the whole fold; it was withdrawn and re-run the same day (A23).

A.6 H6: phase representation. Preregistered as exploratory (A18). The phase block (rank phases, train-only whitening, after Mancilla and Tagliani) is given to every arm; the reported quantity is the shift in the quantum-minus-classical delta between representations. Ten ULB seeds, pairs over every column (30 baseline, 90 with phases: 435 and 4,005 variables, built over all columns because a top-13 selection would exclude every phase column), CVQBoost via the exact proxy [SIM], zero metered seconds.

Mean delta: -0.0685 in the baseline representation, -0.0800 with phases; representation shift -0.0115, paired seed-to-seed SD 0.0135, negative on 7 of 10 seeds. The shift is 0.85 of its paired SD and 43% of the MDE; the preregistered falsifier ($|\text{shift}|$ below the paired SD) fires. Baseline mean AP: XGBoost 0.8328, CatBoost 0.8185, trained-frequency GAM 0.7893, CVQBoost 0.7646; a GBDT set the bar in all 20 cells, so the twins changed what “best classical” means, not the number (GBDT figures sit below A.1 because every arm takes a fixed 30-column budget). A twin-budget handicap under the phase block can only raise the delta; the shift comes from CVQBoost falling further (0.7646 to 0.7513) than the GBDT bar (0.8331 to 0.8313).

B. Preregistration registry

B.1 Gates, scored as committed.

Gate and criterion	Outcome	Tag
G0: tuned XGBoost mean AUPRC ≥ 0.85	FAIL: 0.8296; halt rule not honoured, deviation A28	SIM
G0 leakage tripwire: no cell above 0.95	PASS: 0.8368	SIM
Shuffled-label tripwire: collapses to base rate	PASS: 0.0023 / 0.0017	SIM
G0b: proxy vs hardware Spearman ≥ 0.5	PASS: 0.900	HW
H1b (primary): CVQBoost vs matched GBDT	NULL: -0.0399, interval excludes zero	HW
H4: vs best structural control	PARTIAL: solver fidelity only; committed controls unrun	HW
H3: feature ladder closes the gap?	NO: slope -0.006 per feature; hardware within 0.0013 of proxy	SIM+HW
H6: phase representation moves the delta?	NO: shift -0.0115, falsifier fired	SIM
H1a, H1c, H5, cardinality arm	NOT RUN (vs MIQP, greedy, annealing controls)	PROJ

H1b is the sole confirmatory endpoint, reported unadjusted; all else is exploratory. The compound falsification criterion (H1b null, H3 slope non-positive, H5 non-transfer) has two of three conditions fired and H5 unrun.

B.2 Amendments. Thirty-two dated amendments, A1 to A32, each with rationale and approval, in the repository. Cited here: A11 to A18 register exploratory arms; A12 the free-tier 100-variable ceiling; A17 every A11/A13 figure recomputed after three fold builders skipped deduplication, moving the tuned-pool result from below to above the MDE; A20 memorisation, not majority-class collapse; A21 the ceiling was a billing tier; A22/A23 the hardware ladder published, withdrawn, re-run; A26 the comparator certified by KKT residual; A27 two mechanism claims withdrawn; A28 two protocol deviations including the G0 halt; A31 frozen-pool agreement forced by resolution; A32 the linear-term spread in A31 corrected. No amendment changed a gate criterion; no gate was rescored after observation.

B.3 B2, the largest metered configuration. Top-17 features, order-3 pool, 833 continuous variables against B1’s 91; foreclosed by A12 until A21, run on 2026-09-10. Eleven fits, 906 metered seconds: ten stratified seeds plus the temporal protocol. Ten-seed means: B2 (833 variables) 0.7928 AUPRC (SD 0.0296), AUC-ROC 0.9423 [HW]; B1 frozen (91 variables) 0.7671 [HW]; CatBoost, all features, 0.8368 [SIM]. Temporal protocol: 0.7605 AUPRC, 0.9261 AUC-ROC [HW]. Paired on identical seeds, B2 minus B1 is +0.0256, 95% CI [+0.0203, +0.0310], 10 of 10 seeds (unpaired it hides inside a seed SD of 0.028). k (13 to 17) and subset order (2 to 3; three-feature learners are 680 of the 833 variables) moved together, so a larger, richer pool scores better; the ceiling alone is not shown to have cost 0.0256. The disconfirming cell, k=17 at order 2 (153 variables), is Phase 2 experiment 1. The objective stays convex: a three-feature tree is still one variable emitting a vector in $\{-1, +1\}$, and on B2’s own pool the smallest eigenvalue of $J = HH^T + \lambda I$ is exactly λ (A27). The device solves this block less faithfully: weight cosine 0.8192 to 0.8327 (mean 0.8272) against 0.9747 to 0.9825 for order-2 cells, and the recomputed objective sits 191.5 above the classical minimum. The cause is resolution: at sum constraint 1 the device resolves about 0.005 per weight, a diffuse optimum over 833 learners averages 0.0012, every fit returned exact zeros, and printed nonzero weights run 0.0007 to 0.0029. The 0.83 cosine measures a device solving a sparsified version of the problem, which is why cardinality-constrained selection is the Phase 2 direction. Score health: mode share 0.857 over 4,412 distinct values (B1: 0.951 over 814); the tie block sits at the bottom of the range, so AUPRC ambiguity is 0.0009 against 0.0804 for AUC-ROC, and only AUPRC is sound here.

C. References. (1) Loke et al., Improving Credit Card Transaction Fraud Detection Using CVQBoosting, ICAART 2026, DOI 10.5220/0014628400004052. (2) Emami et al., Financial Fraud Detection with Entropy Computing, arXiv:2503.11273. (3) Nguyen et al., Entropy computing, a paradigm for optimization in open photonic systems, Communications Physics 8, 411 (2025), arXiv:2407.04512; critical comment arXiv:2605.03612. (4) Mancilla and Tagliani, The Fourier Wall, arXiv:2607.15815. (5) Deotte, XGB Fraud with Magic, Kaggle (2019); Yakovlev and Deotte, IEEE-CIS 1st place write-up (2019). (6) FCA FG22/5, Consumer Duty guidance (2022), paras 5.12, 1.27 to 1.28, chapter 11; Equality Act 2010. (7) QCi Dirac-3 documentation. Datasets: MLG-ULB Credit Card Fraud Detection; IEEE-CIS Fraud Detection (Kaggle).